



VISHWANATH KARUNANITHI

Electronics and Communication Engineer

PROFILE SUMMARY

Electronics and Communication Engineering student with strong hands-on experience in embedded systems, Internet of Things, software-based systems, and artificial intelligence applications. Actively involved in real-world project development, hackathons, internships, and innovation challenges. Passionate about creating offline AI solutions and Digital Twin technologies.

EDUCATION

Bachelor of Engineering in ECE

Velammal Institute of Technology, Chennai

2023 – 2027 (Pursuing) | CGPA: 8.47

Higher Secondary Education (CBSE)

Cauvery International School, Tiruchirapalli | 80.60%

Secondary School Education (CBSE)

Cauvery International School, Tiruchirapalli | 81.80%

MAJOR PROJECTS

Bio-Guardian: Smart Non-Invasive Health Monitoring

Implemented a non-invasive system for continuous monitoring of vital health parameters. Showcased at Vadiva Tech Fest (TVIS) for three days.

Piracy Prevention and Detection

Designed a dual-layer system using IR integration and software-based detection to prevent unauthorized recording of digital content.

Structural Health Monitoring (Software)

Developed a software-based system to analyze structural parameters and assist in early fault detection.

Electricity Tampering Detection and Smart Energy Meter

Built an embedded IoT-based system for real-time energy monitoring and electricity theft detection.

Exam Hall Allocation System (Embedded)

Designed a microcontroller-based automated exam hall seat allocation system.

Face Detection Using Artificial Intelligence

Developed an AI-based computer vision system for real-time face detection.

ACHIEVEMENTS & EXPOSURE

Participant – 24-Hour Hackathon at Sir M. Visvesvaraya Institute of Technology, Bangalore. Gained extensive experience in rapid prototyping and teamwork.

CONTACT

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vishwanath-karunanithi-087b62308

GitHub

github.com/

vishwanathkarunanithi

Portfolio

vishwanathkarunanithi.

github.io

TECHNICAL SKILLS

Embedded Systems

Internet of Things (IoT)

Arduino and ESP32

STM32 (Blackpill & Bluepill)

Sensors and Microcontrollers

AI & Computer Vision (Basic)

Digital Twin Technology

Python (Basic)

TOOLS AND PLATFORMS

Arduino IDE

MIT App Inventor

LABVIEW

Sinric Pro

Hugging Face

GitHub

LANGUAGES

English

Tamil

Project Showcase – Vadiva Tech Fest at The Velammal International School (TVIS). Selected to display the "Bio-Guardian" project for three consecutive days.

Conference Attendee – Jerusalem College of Engineering, participated in technical sessions and knowledge exchange.

Finalist – YUKTI Innovation Challenge 2025, selected for the final stage based on innovation potential and technical feasibility.

First Prize – Inter College Project Expo for an embedded and software-based project evaluated by academic and industry professionals.

Winner – 24 Hour Hackathon at Vellore Institute of Technology, involving rapid problem solving and system implementation.

Internship – One Yes Info Tech, completed an EEPROM-based embedded project.

TN Skills Competition, selected up to Round 2.

Three First Prizes – AMS College of Engineering across technical events.

Industrial Visits to IIT Communication Laboratory and RETECH.

MINI PROJECTS AND TECHNICAL ACTIVITIES

Offline AI Heart Monitoring (Digital Twin): Developed an AI model utilizing Digital Twin technology to monitor heart health. The system is designed to function completely offline without internet connectivity.

Offline AI Coding Assistant: Created a generative AI tool designed to assist with coding tasks. Engineered to operate locally, ensuring functionality without the need for an internet connection.

Hugging Face Exploration: Explored and implemented various open-source models from Hugging Face for Natural Language Processing (NLP) and Computer Vision tasks.

STM32 Development: Worked with STM32 Blackpill and Bluepill boards using the Arduino framework for advanced microcontroller applications.

HVAC Controller: Developed a Heating, Ventilation, and Air Conditioning controller system.

Otto Blockly Robot: Mobile application-based customization, home automation using Alexa and Sinric Pro, Bluetooth-controlled LED using MIT App Inventor, and completion of all basic LABVIEW exercises.

DECLARATION

I hereby declare that the details furnished above are true and correct to the best of my knowledge and belief.

Date: _____

Place: _____

Signature